

Fig. 2: Actual-size PCB layout of bi-directional DC motor control

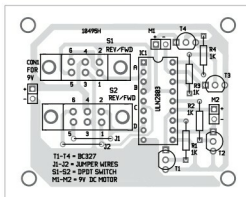


Fig. 3: Components layout for the PCB

Case 3 (M2). When point C = 0 and D = 1, output pins 13 and 14 of IC1 will be high (9V), and pins 11 and 12 will be active-low. Transistor T2 is in cut-off state and transistor T1 conducts. Motor M2 rotates in one direction, say, forward direction.

Case 4 (M2). Similarly, when point C = 1 and D = 0, output pins 13 and 14 of IC1 will be low, and pins 11 and 12 will be active-high (9 volts).

Transistor T2 conducts and T1 is in cut-off state. Motor M2 rotates in the opposite direction.

If all the inputs (A, B, C and D) are at logic 0 (GND), both the motors will be in stop condition because there will be no voltage differences at the motors' terminals.

Note. If all the inputs are at logic 1 (9 volts), transistors will burn out. So it is not recommended to apply

both the input terminals with the same high voltages in this circuit configuration.

Construction and testing

An actual-size PCB layout of the bi-directional DC motor control circuit is shown in Fig. 2 and its components layout in Fig. 3. Assemble components on the PCB. Connect motors at M1 and M2 markings

on the PCB. Connect a 9V PP3 battery for power supply to the circuit. You can use double-pole double-throw or slide switches for S1 and S2. **EFY**



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